

Knowledge Spillovers Within Regional Networks of Innovation and the Contribution Made by Public Research

Martina Kauffeld-Monz*

Several studies have highlighted the importance of regional innovation networks concerning localized knowledge spillovers. Furthermore, the contribution of public research to innovation, as well as its role within regional innovation systems, has been discussed in general in the existing literature. This paper, in an attempt to link up these ideas, investigates knowledge transfer processes within inter-organizational R&D projects that are embedded in 23 German regional innovation networks (with almost 500 participants). The first part of the paper empirically focuses on the determinants of knowledge spillovers within these regional innovation networks. The results show strong ties, network cohesion as well as cooperation experience as crucial factors regarding the acquisition of new knowledge. Moreover, it is found that knowledge exchange is increased by the degree of specialization that results from network participation. Furthermore, empirical results show universities' prominent role in knowledge transfer within the inter-organizational processes of innovation, when compared to other groups of actors distinguished by the regional innovation systems approach. Finally, the results support the assumption that public research has an 'antenna function' (boundary-spanning function) for the firms involved in the networks, due to its integration into the international scientific community. Manufacturing firms can be seen as the (net) winners, with respect to the knowledge exchange processes.

Introduction

Innovation process networks have a superior mechanism of coordination, when compared to coordination by hierarchy or by markets (Powell, 1990). They are:

- The dynamics of technological progress, technological complexity and the rising costs of R&D result in a need to incorporate resources and capacities external to the firm, these being of much greater importance to small firms than to larger ones (Link and Rees, 1990; Acs *et al.*, 1994; and Audretsch and Vivarelli, 1994). Becker and Dietz (2004) found that sources external to a firm, attracted by R&D-collaboration, influence the probability of the development of new products. They also found that the likelihood of realizing product innovation rises with the number of parties involved in cooperation.
- Where innovative division of labor by markets has to be coordinated, there are special problems regarding knowledge transfer, in the context of innovation activities. These are incomplete contracts, 'thin markets' for innovative inputs, and a risk of unintended knowledge spillovers (Fritsch, 2001a and 2001b). Beyond this, in the innovation process, much uncoded or 'tacit knowledge' (Polanyi, 1967) must frequently be

* Dipl.-Ökonomin, Institute for Urban Research and Structural Policy, (IfS Berlin), Mainzer Str. 18, 10715 Berlin, Germany. E-mail: kauffeld-monz@ifsberlin.de

exchanged—knowledge that is difficult to codify, and which, if generated in the private sector, is often too specialized for publication. Therefore, personal contacts are necessary for the acquisition of this kind of knowledge, face-to-face contact being especially helpful (Von Hippel, 1994), at least in the beginning of a new collaboration. Furthermore, intensive face-to-face-contacts are an efficient way to accelerate the creation of trust—something that is indispensable in the exchange of sensitive firm information and knowledge, which are critical to the core competencies of a firm.

Besides cultural or social proximity (Breschi and Lissoni, 2003) and spatial proximity (Fritsch, 2003), these above conditions explain why the cooperation of partners is beneficial to the process of knowledge exchange. This is one part of the phenomenon whereby innovation processes have a pronounced regional dimension (Feldman, 1994; Scott, 1996; Breschi, 1998; Beise and Stahl, 1999; Boekema *et al.*, 2001; and Fritsch *et al.*, 2007).

Networks involve more than the actual bilateral or multilateral cooperative projects—networks offer an indefinite quantity of possible relations between organizations and actors not yet working together. Since it is often very difficult and expensive to find suitable new partnership, networks involved options of access to a pool of trustworthy potential new partners. Moreover, network embedded R&D partnerships foster trust since opportunistic behavior damages one's reputation within a network (Gulati and Gargiulo, 1999). Thus, 'governance by control' may be replaced by 'governance by trust', which makes networks instrumental concerning the reduction of transaction costs (Nooteboom, 1999; Bartelt, 2002; Dyer and Chu, 2003; and Nooteboom and Gilsing, 2004). Trust-based network relations not only involve possibilities of reduction of transaction costs, but also increase transaction value, such as the exchange of intangible resources like knowledge (Zaheer *et al.*, 1998; Nooteboom, 1999; Nooteboom, 2002; and Dyer and Chu 2003). Furthermore, the importance of public research organizations, within these regional networks of innovation, is emphasized (Fritsch and Schwirten, 1999; Revilla Diez, 2000; Fritsch, 2001a and 2001b; and Varga, 2006). Thus, innovation networks at the regional level, which involve both firms and public research organizations, are seen as crucial elements of proper functioning of regional innovation systems (Fritsch and Slavtchev, 2007; and Fritsch and Slavtchev, 2008). But, to what extent do certain characteristics of a network, as well as those of its participating organizations, foster knowledge transfer, remains unclear. Moreover, from the existing literature, one cannot conclude the relative importance of public research organizations in regional innovation networks, with respect to knowledge transfer.

In this study, the determinants of knowledge transfer and acquisition between actors of inter-organizational R&D projects, which are embedded in regional innovation networks, are analyzed, whereby special attention is given to the contribution of public research. The paper first gives a brief insight into the relevance of public research with respect to innovation. Then, it outlines the focus of the research, lays out some hypotheses, and introduces the database. Finally, it presents the results, before concluding.

Science-Business Links

It is hardly surprising that knowledge flow between science and business increased threefold in the US between the end of the 1980s and the mid 1990s (Narin *et al.*, 1997). The increased relevance of science-business indirect links, as well as direct interactions, is based on both the multidisciplinary character of research and a stronger dependence on technological advances of science, which is also becoming apparent in the increasing role of R&D due to the generation of new products (Grupp and Schmoch, 1992; and Rosenberg and Nelson, 1994). In addition, policy measures with regard to the transfer of knowledge from science to business have enormously affected the links between the public and private spheres (Hall *et al.*, 2000; Cohen *et al.*, 2002; Fontana *et al.*, 2004; and Lawton-Smith, 2007).

Results of the analysis based on the knowledge production function approach confirm the relevance of public research to private innovation (Jaffe, 1989; Acs *et al.*, 1992; Feldman, 1994; Blind and Grupp, 1999; and Becker, 2003). Yet, this kind of analysis based on the aggregate data does not provide information on how this scientific knowledge is channelled into the economy or who the agents and recipients of this knowledge are. From the surveys, we know what kind of firms are cooperating with public research—they are usually large companies which operate their own R&D departments (Fritsch and Lukas, 2001; Mohnen and Hoareau, 2002; Laursen and Salter, 2003; Busom and Fernández-Ribas, 2004; and Fontana *et al.*, 2004). The number of R&D collaborations with public research, however, is affected by the relative size measured by R&D employment (Fontana *et al.*, 2004). These findings result from a certain level of ‘absorptive capacity’ that is necessary if the value of new information and knowledge is to be recognized, assimilated and applied for commercial ends (Cohen and Levinthal, 1990).

Hence, concerning a majority of firms, public research organizations are not seen as the most important sources during the innovation process, neither at the ideas stage nor at the completion phase of innovation (Cohen *et al.*, 2002; and Fontana *et al.*, 2004). Nevertheless, many innovations—if they had gone ahead at all—would not have been realized without close collaboration with public science (Mansfield, 1991; Mansfield and Lee, 1996; and Beise and Stahl, 1999). So, it is not surprising, as Broström and Lööf (2008) point out that for Swedish firms, knowledge transmission between universities and industries has a positive affect on both innovative input and innovative output (measured by the propensity to apply for a patent and the amount of innovation sales per employee).

It has been demonstrated that many achievements of public research establishments can be attributed to their immediate, spatial surroundings (Jaffe 1989; Acs *et al.*, 1992; Anselin *et al.*, 1997; Beise and Stahl, 1999; Varga, 2000a and 2000b; Autant-Bernard, 2001; Fritsch and Schwirten, 2000; Varga, 2006; and Fritsch and Slavtchev, 2007)¹, although spatial proximity turns out to be less important for basic research than it is for applied R&D (Mansfield, 1991 and 1995). Hence, most of the private sector cooperation partners of universities are located at a relatively close distance (Fritsch, 2003 and 2005).

¹ That is impressively illustrated by famous case studies: Silicon Valley, Route 128, and Cambridge Phenomenon.

University-industry partnerships especially stimulate new ideas and are instrumental in creating and bringing radical innovations to the market (Belderbos *et al.*, 2004). But the processes in which the universities are involved are also directed towards completing the existing R&D projects (Cohen *et al.*, 2002). Focus on one or the other activity differs depending on the socioeconomic structure of a region and the quality of regional innovation system in which the public research organization is embedded (Fritsch, 2004). Boucher *et al.* (2003) analyzed the degree of university involvement in their location considering the regional as well as universities' characteristics and came to the following conclusions: Universities serving a peripheral region as a single-player research organization have the highest degree of involvement in their location²; secondly, the competition between universities for local institutional dominance reduces their involvement in the region (a situation often found in urban regions). Finally, traditional universities feel more compelled to maintain their position within the international hierarchy than to cultivate their activities in their own locality. Furthermore, Fritsch and Slavtchev (2007) provide empirical evidence that the intensity and quality of the research conducted by the universities have a significant effect on the regional innovative output while pure size of a university is unimportant. But even the best of publicly-funded science is only of benefit to the location if it is spatially embedded in a well developed economic environment (Varga, 2000a and 2000b). Varga's (2000a and 2000b) study demonstrated that identical amounts of university research spending can be associated with dramatically different levels of innovation output, depending on the concentration of economic activities.³ The fact that a dominating firm in a regional innovation network is losing its central position to the local university (with respect to patents) may be regarded as an extreme of this, but may not be untypical of the transformation process in the eastern part of Germany (Cantner and Graf, 2004).

With regard to the role of public research in the regional innovation systems, one can identify at least four functions (Varga, 2000a and 2000b; Gunasekara, 2006; Fritsch *et al.*, 2007; Fritsch and Slavtchev, 2007; and Lawton-Smith, 2007): (i) generation of new knowledge; (ii) accumulation of this knowledge and the knowledge originating elsewhere; (iii) transmission and transfer of all accumulated knowledge. Based on the discussions of the changing modes of knowledge production (model 2) and 'academic entrepreneurship' as well as 'the entrepreneurial university' (Gibbons *et al.*, 1994; Etzkowitz and Leydesdorff, 2000; and Etzkowitz *et al.*, 2000)⁴ a further function is increasingly attributed to public research; (iv) the conversion of research results in innovation: public research organizations and especially universities may serve as an incubator for knowledge intensive spin-offs (Stankiewicz, 1994; Jones-Evans, 1998; Lockett *et al.*, 2003; Fritsch and Slavtchev, 2007; and O'Gorman *et al.*, 2008). Moreover, public research organizations sometimes play a major role in the location decisions of these new firms, especially in the case of scientific spin-offs, since

² However, these are the regions that suffer most of all from graduate migration to the more developed regions.

³ These findings correspond with what Feldman (1994) suggested in her study of the regional effects of John Hopkins University: no 'critical mass' of high-tech companies, lack of service providers, lack of entrepreneurial culture and venture capital.

⁴ The increasingly networked mode of knowledge production, in which the rigid borders between disciplines, interest positions and groups of participants dissolve (Etzkowitz and Leydesdorff, 2000); Gibbons and Limoges *et al.* (1994).

proximity to the incubator is important for the firms' development, at least in the initial phase of development. However, this proximity usually loses its importance over time (Lindholm-Dahlstrand, 1999). These general functions of public research organizations also become effective within networks of innovation (Lambooy, 2004).

A lot of analyses are made of the ways in which knowledge stemming from public research spills over to business (Schmoch, 1999; Agrawal, 2001; Schartinger *et al.*, 2002; and Zellner, 2003). The channels of transfer range from publications—a very impersonal channel—and collaborative research to the migration of scientists to established firms or via spin-off. But the knowledge about the relative importance of several channels as well as their mutual interdependence is very fragmented (Fritsch and Schwirten, 1999; Fritsch, 2001a and 2001b; and Fritsch *et al.*, 2007). For example, Cohen *et al.* (1998), Cohen *et al.* (2002) and Meyer-Kramer and Schmoch (1998) found the exploitation of certain channels varying considerably across industries. However, a number of empirical studies provide evidence that R&D cooperation can be seen as a relatively effective means of knowledge transfer from science to business (Fritsch and Schwirten, 1999; Fritsch and Lukas, 2001; Monjon and Waelbroeck, 2003; Arundel and Geuna, 2004; and Fritsch and Franke, 2004). Typically, R&D-intensive firms conduct R&D with the cooperation of public research (Link and Rees, 1990; Beise and Stahl, 1999; Fritsch and Lukas, 2001; Mohnen and Hoareau, 2002; and Laursen and Salter, 2004). But it has also been shown that the firms' R&D capacity is not sufficient regarding the acquisition of scientific knowledge. The rate of absorption especially results from the degree of interconnectedness with public research organizations (Cockburn and Henderson, 1998; Lim, 2000; and Schmidt, 2005). Thus, regional innovation networks can be seen as facilitating the acquisition and diffusion of science-based knowledge.

Design of the Analysis and Database

Research Agenda and Hypotheses

With respect to knowledge and innovation management, one can draw the following questions from the sections above: (1) Is participation in regional innovations networks a promising endeavor concerning the access to science-based knowledge? (2) Which are the general conditions that support knowledge transfer with respect to certain network features as well as to the characteristics of the network participants?

The Question (2) is addressed at first. A regression analysis is conducted to identify factors that influence the degree of knowledge absorption in the course of inter-organizational R&D projects that are conducted by participants of regional innovation networks. Although strong ties involve the danger of cognitive 'lock-in' processes under certain circumstances (Granovetter, 1973; and Grabher, 1993), in this study, it is supposed that strong ties positively effect the knowledge exchange in an early stage of emerging innovation networks. Ties have to be strong because a considerable amount of sensitive information and knowledge that is critical to the core competences of a firm has to be exchanged. Moreover, the exchange of tacit knowledge needs not only personal contacts but also frequent face-to-face interaction that involves trust.

H₁: It is assumed that the contact intensity and trust based strong ties foster the absorption of knowledge.

But networks typically involve more than trust-based dyadic ties.

H₂: Often, the interaction within networks leads to the development of norms of conduct and to a certain degree of mutual identification (network cohesion) that may reduce the danger of opportunistic behavior resulting from information asymmetries.

Furthermore, it is assumed that it is more efficient to have some experience in handling such networks (for the most part of tacit knowledge) on account of the range of knowledge one can extract.

H₃: Thus, the existence of network or R&D cooperation experience may be helpful concerning the acquisition of knowledge.

The degree of knowledge absorption is measured on a 5-point Likert scale as well as the tie strength (trust) and the two indicators of network cohesion (Appendix). Moreover, the analysis takes into consideration that these relations may be different for certain groups of actors that are involved on the regional innovation networks. The firms, for example, may tend to strongly rely on trust-based relations because they are faced with greater economic risk and relational risk with respect to inter-organizational innovation processes. Thus, additional regression analyses are conducted for the group of the firms as well as the group of public research organizations. Small firms may be restricted in knowledge acquisition due to their 'absorptive capacity' (Cohen and Levinthal, 1990).

Furthermore, this paper argues that intensive contacts are crucial not only to the exchange of (tacit) knowledge (Noteboom, 2001), but also to the screening and transfer of relevant information. For the information to be really useful to the recipient, the information provider must know exactly what the needs of the recipient are. The usefulness of the received information thus depends on the intensity of the contact between the information transmitters and receivers. There will not necessarily be an increase in the amount of information exchanged, but it will be more purposeful.

H₄: Hence, intensive contacts are crucial not only for the acquisition of (tacit) knowledge but also for the absorption of information, measured on a 5-point Likert scale (Appendix).

H₅: It is assumed that the degree of labor division that results from the participation in a regional innovation network in turn gives way to the extension of the knowledge exchanged.

In the second part of the paper, the relevance of public research organizations to the process of knowledge exchange is analyzed.

H₆: Universities and non-university research organizations contribute to knowledge exchange and benefit from it

On the one hand there the extent to which universities and non-university research organizations contribute to knowledge exchange and benefit from it (*H₆*), and on the other

hand, the ‘antenna function’ (Fritsch and Schwirten, 1998; and Revilla-Diez, 2002) of public research that results from its integration into the international scientific community is being examined empirically.

H₇: It is supposed that the public research organizations’ frequency of interregional R&D cooperation (Appendix) is positively related to the amount of knowledge they transfer to network partners.

This aspect is of crucial importance for (small) firms that are by themselves not well linked to knowledge sources external to the region and thus, are restricted concerning the acquisition of new knowledge that is diverse to the regional knowledge base.

Data

As a way of strengthening the innovation system and reducing what is seen as one of the weaknesses of the new post-1990 federal states of Germany,⁵ the InnoRegio-Program was launched in 2000 by the German Federal Ministry for Education and Research (BMBF) in order to stimulate the formation of regional innovation networks. Not limited to certain technologies or industries, the BMBF is set to provide approx. € 250 mn of support to 23 contest selected networks until 2006 (Eickelpasch *et al.*, 2002a and 2002b; BMBF, 2005; and Eickelpasch and Fritsch, 2005). The financial support was addressed to inter-organizational R&D cooperation especially with focus on the collaboration between public research organizations and Small and Medium sized Enterprises (SMEs). The networks are very heterogeneous concerning their goals, and the industries and technologies involved (Biotechnology, medical technology), and owing to the respective number of participants (Eickelpasch *et al.*, 2001). Commonly, it is in the networks that participants from public research and business located in the same region (with a maximum of 50 km) collaborate with one another. Moreover, the regions involved suffer from similar disadvantages such as low income and productivity, lack of larger firms, etc., that are largely a result of the transformation process in the East Germany, the former German Democratic Republic (Kronthaler, 2005). For these reasons, the networks should be highly comparable.

The innovation initiative InnoRegio was accompanied for over five years by an evaluation team (Kauffeld and Wurzel, 2003; and BMBF, 2005). The data that was gathered by postal questionnaires in the year 2004 resulted in a rather high response rate of about 80% (approximately 500 participants). The data contained information about the performance of the firms, the extent to which information and knowledge were exchanged, network cohesion, strength of ties and much more. The differentiation of the surveyed groups of participants was similar to that of the regional innovation systems approach, transferring of the results to regional innovation systems possible. About 60% of the organizations were private firms. Universities amount to 10% of the total, and about 16% of the actors were public or private non-university research institutes. Most of the firms involved are SMEs with 50% having less than 20 employees and only 10% having more than 100 employees. The service sector firms, which contribute to about 40% of the private firms in the networks, are mainly engaged

⁵ For example, innovation efficiency, DIW, IfW *et al.* (2003), “Zweiter Fortschrittsbericht wirtschaftswissenschaftlicher Institute über die wirtschaftliche Entwicklung in Ostdeutschland”, DIW-Wochenbericht (47/03).

in engineering services and R&D. The manufacturing firms include a high proportion of control technology, mechanical, medical and measurement engineering (Eickelpasch *et al.*, 2002b). The firms in the networks exhibit an above average performance with regard to R&D and introduction of new products in the market, and they consider themselves to be more competitive than most of the other firms in the respective market (Eickelpasch *et al.*, 2002b). For this reason, there is a certain sample selection bias with regard to innovation attitudes, innovative capabilities as well as expectations about future growth. However, many of the firms do not exhibit interregional R&D cooperation and thus, may lack access to knowledge diverse to their regional knowledge base. Moreover, only 30% of the networks' participants had experience of regional innovation networks even though about 60% collaborate in R&D.

Results

Network Experience, Tie Strength, and Network Cohesion

Model 1 points out that network experience has no significant effect on the extent of knowledge absorbed by the firms from their network partners. Network experience is found to be advantageous for public research organizations (Model 2). With respect to the firms, it is particularly the experience collaborating with process innovations that increases the extent of their knowledge absorption. However, cooperation experience in basic research tends to decrease the firms' extent of knowledge they acquire from their network partners. Probably, firms with rich cooperation experience in basic research rely more on other sources of knowledge than on their regional innovation networks. However, on the whole it is shown that previous network or R&D cooperation experiences involve benefits with regard to knowledge acquisition if the actors participate in regional innovation networks at a later point of time (H_3) (Table 1).

The results show that the intensity of contacts as well as the strength of ties (measured by the level of trust) are positively related to the extent of knowledge acquisition (Model 1). Whereas, the public research organizations' extent of knowledge acquisition is based on the R&D partners' intensity of contacts and the knowledge acquisition of firms turns out to be very sensitive to their R&D partnerships' level of trust. The intensity of contacts plays a minor role in the firms' absorption of knowledge. Moreover, tie strength (as indicated by trust) is found to be the most important aspect with respect to the firms' acquisition of knowledge. In this respect, the findings support the H_1 i.e., strong ties can be seen as being an important factor in knowledge exchange at the initial stages of innovation networks.

The variables 'norms' (COM_NORMS) and 'identification' (IDENT) refer to the characteristics of a network and are seen as indicators for the degree of network cohesion. Differences are again found between the firms and the public research organizations. From the firms' perspective (Model 2), the development of common norms in their network plays a crucial role with respect to knowledge absorption. With regard to public research organizations (Model 3), common norms in the network obviously do not play a significant role. But for both groups, it is shown that the degree of network partners' identification with the network, positively influences the extent of knowledge exchange (Models 2 and 3). These findings support the H_2

Table 1: Regression Summary: Degree of Knowledge Absorbed from Network Partners

Independent Variables	Firms and Public Research Organizations		Firms		Public Research Organizations	
	Model 1		Model 2		Model 3	
Constant	0.971	–	0.423	–	1.138	–
NW_EXP	0.60	(1.465)	0.013	(0.244)	0.140**	(2.138)
CO_EXPBR	–0.084**	(–2.0)	–0.139**	(–2.486)	0.037	(0.566)
CO_EXPPDInno	0.029	(0.624)	0.057	(0.954)	0.127	(1.582)
CO_EXPPCInno	0.112**	(2.479)	0.206***	(3.651)	–0.003	(–0.042)
CON_INT	0.155***	(3.259)	0.012	(0.183)	0.264***	(3.491)
TRUST	0.152***	(3.036)	0.229***	(3.294)	0.018	(0.228)
COM_NORMS	0.092*	(1.861)	0.154**	(2.413)	–0.036	(–0.441)
IDENT	0.155***	(3.181)	0.142**	(2.105)	0.236***	(3.157)
FIRM_SIZE	–	–	0.135**	(2.453)	–	–
Number of observations	492		257		197	
R ²	0.228		0.312		0.263	
Note: * indicates significance at 10% level; ** indicates significance at 5% level; *** indicates significance at 1% level; t-value in brackets.						

that network cohesion contributes positively to the knowledge that is exchanged between the network partners. As supposed, the absorption of knowledge is related to the respective absorptive capacity of a firm that was measured by firm size (Model 2) (Table 1).

Contact Intensity and Information Exchange

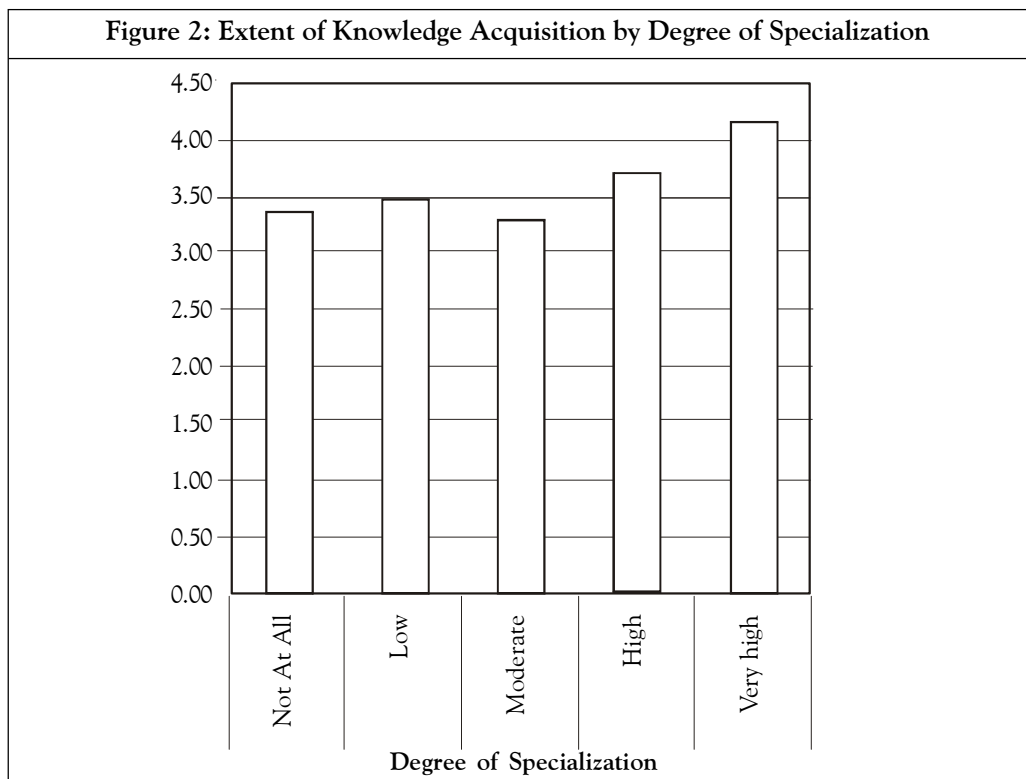
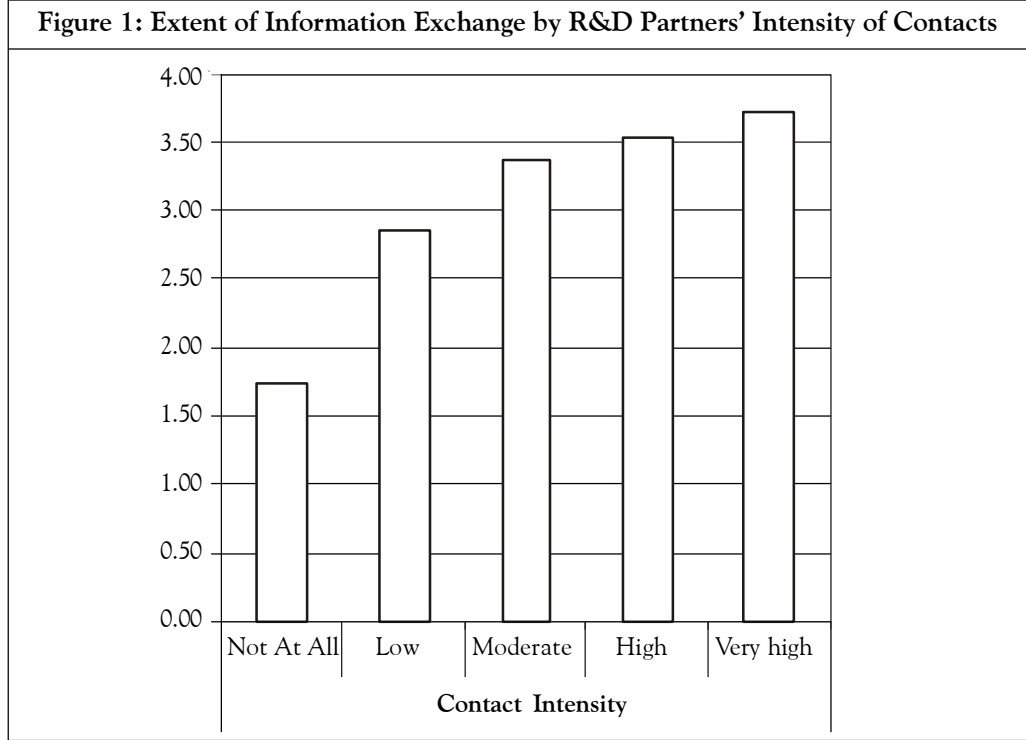
The data allows for a distinction to be made between knowledge and information. Further, following H_4 , strong evidence is found for a positive relation between information exchange and intensity of contacts (Figure 1).⁶ Thus, it can be said that information exchange requires intensive contact just as the exchange of (tacit) knowledge does.⁷

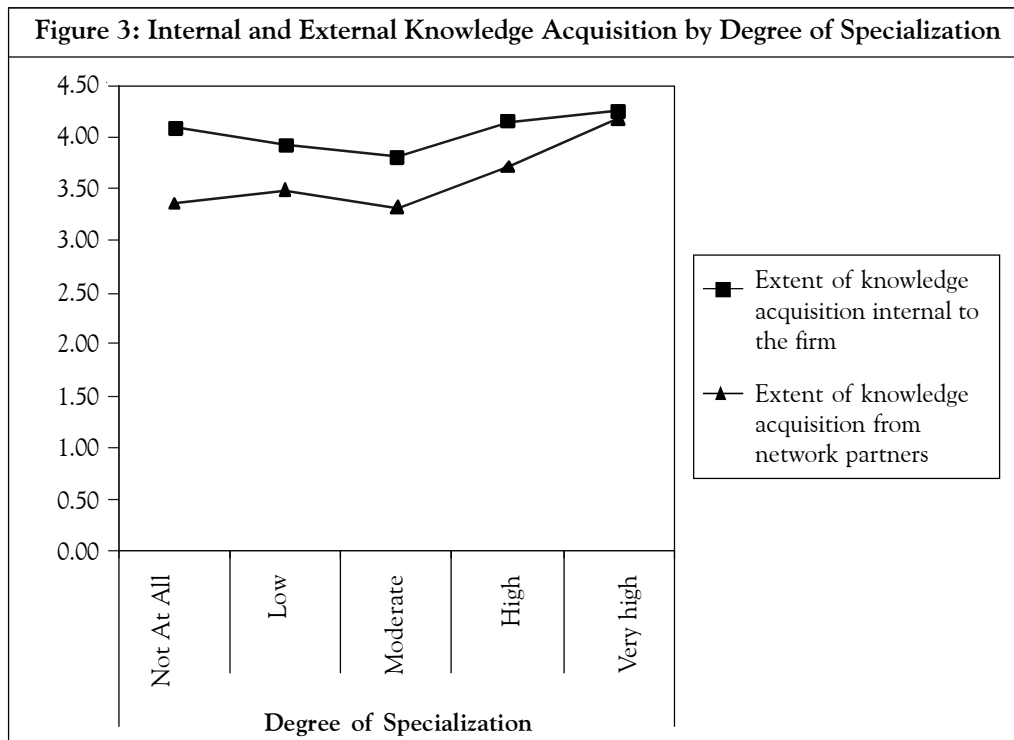
Degree of Labor Division and Knowledge Exchange

The results further indicate (Figure 2), the assumption that increasing division of labor leads to a rise in knowledge exchange (H_5). Moreover, substitution between the firms' internal

⁶ Results of a Mann-Whitney test show that the differences between the moderate/high/very high contact intensity are significant at 1%-level. With respect to 'low' contact intensity (25 cases) as well as 'not at all' (2 cases) contacts analysis is not feasible because of small case numbers.

⁷ But the correlation between contact intensity and knowledge exchange is somewhat stronger (Kendall-Tau-b correlation coefficient of 0.284***) than correlation between contact intensity and information exchange (Kendall-Tau-b correlation coefficient of 0.197***).

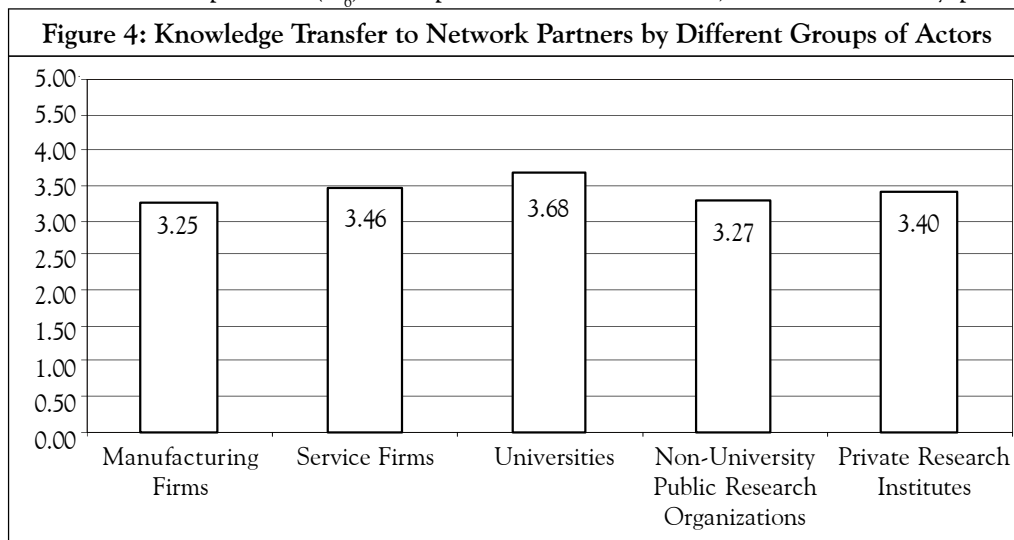




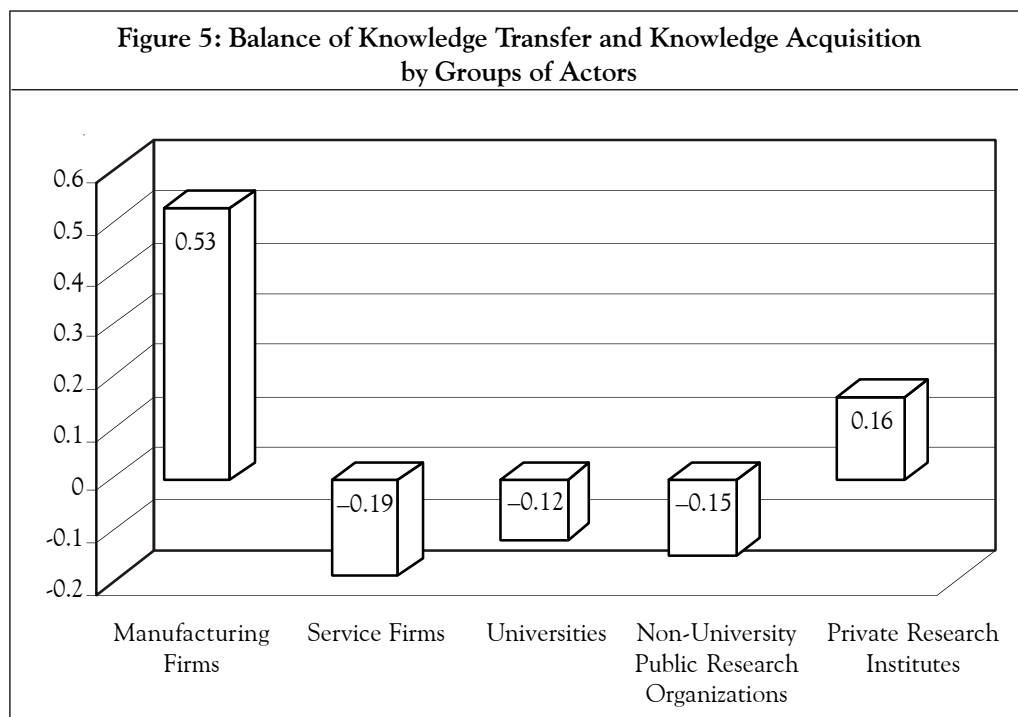
knowledge and the knowledge acquired external to the firm takes up a certain degree of specialization (very high) (Figure 3). The knowledge acquired external to the firm is almost as extensive as the knowledge internal to the firm.

The Contribution of Public Research to the Knowledge Exchange in the Networks

Figure 4 shows that it is the group of universities which transfer the highest extent of knowledge to their network partners (H_0). Compared to the universities, the non-university public



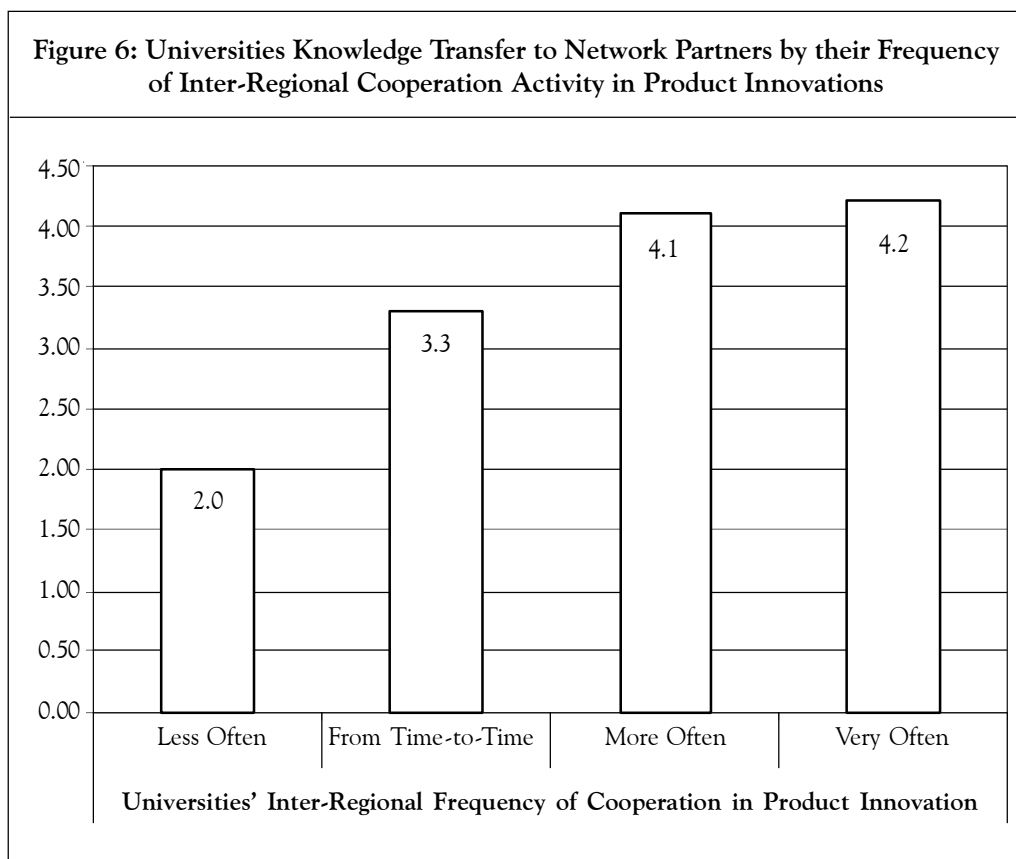
research organizations transfer a significantly lower extent of knowledge to the partner of their respective networks. In consideration of the knowledge transfer as well as the knowledge acquisition, one gets insight with respect to the 'net benefits' of knowledge exchange for the different groups of actors (Figure 5), whereas the manufacturing firms make up the group that gains 'net benefits' in the course of knowledge exchange between universities and the non-university public research organizations. Service firms also transfer knowledge to their network partners, more than what they acquire from them. But, there is another group of actors that attains a positive balance between knowledge transfer and knowledge acquisition: the private research institutes that are a special feature of the research landscape in Germany's new federal states.



The 'Antenna Function' of Public Research

The investigation of the 'antenna function' (H_7) is based on the correlation between an organizations' frequency of interregional cooperation (in basic research, product innovations and process innovations) and its extent of knowledge transfer to networks partners. The findings relating to the 'antenna function' (Table 2) show that in universities the antenna function becomes effective. There is significantly a positive correlation between the universities frequency of inter-regional cooperation activity in product innovations (Figure 6) as well as process innovations and the extent of knowledge they transfer to their network partners (Table 2). With respect to the non-university public research organizations, however, this does not apply (Table 2). But a further group seems to be enormously important to any consideration of the 'antenna function' (Table 2) i.e., the private research institutes.

Table 2: Antenna Function by Groups (Correlation Coefficients)			
Correlation Coefficients Between the Frequency of Inter-Regional Cooperation Activity and the Extent of Knowledge Transfer to Network Partners			
Group of Actors	Basic Research	Product Innovations	Process Innovations
Manufacturing Firms	0.70 (47)	0.105 (72)	0.112 (63)
Service Firms	0.262 (29)	−0.061 (37)	0.009 (37)
Universities	0.057 (64)	0.468*** (47)	0.310* (39)
Non-University Public Research Organizations	0.165 (28)	0.264 (19)	0.227 (20)
Private Research Institutes	−0.007(40)	0.465*** (37)	0.490*** (34)
1 = supra-regional cooperations (yes/no); 2 = in case of 1= yes: frequency of cooperation; (N)			
Note: *indicates significance at 10% level; ** indicates significance at 5% level; *** indicates significance at 1% level.			



Whereas, the antenna function comes out to be effective with regard to product innovations and process innovations, the frequency of interregional cooperation activity in basic research obviously has no influence on the knowledge that is transferred to network partners (Table 2). Probably, the knowledge that is attained in the course of those kinds of cooperation is not as relevant in the innovation processes of the networks under study as the knowledge that is attained in the course of cooperation in the product of process innovations.

Conclusion

Knowledge-based view of the growth of the firm states that knowledge is the essential resource of a firm. In this study, it is examined under which conditions regional innovation networks can be an adequate means for firms to acquire knowledge that they need for innovation but that cannot be generated and provided in-house.

With respect to the firms, the results indicate that in particular, trust-based strong ties positively influence the extent of knowledge acquisition. Furthermore, the development of norms of conduct as well as the network participants' identification with the network is found to be conducive with respect to the knowledge exchange in the network. This implies that investments in relational capital as well as in network cohesion, on the one hand are time consuming and costly, while on the other hand they involve benefits. If these investments economically pay off cannot be answered on the basis of the data. Moreover, it has to be considered that strong ties in the long run involve the risk of cognitive lock-in processes (Granovetter, 1973; and Grabher, 1993) that may inhibit the acquisition of information and knowledge diverse to the firms' knowledge base. Probably, there is an inverted U-shaped relation between tie strength and information and knowledge acquisition over time. Whereas, strong ties are crucial with regard to early stages of R&D partnerships. They turn out to be counterproductive in the advanced stages.

With respect to public research organizations, neither trust-based relations nor norms, are found to be essential for the knowledge exchange, but intensive contacts do. Probably, this refers to the fact that public research organizations in the course of inter-organizational innovation are not as much as the firms faced with such kind of risks as, for example, unintended knowledge spillovers.

Furthermore, the results indicate that innovation network experience and R&D cooperation experience foster the acquisition of knowledge. For this reason innovation policy that is addressed to innovation networks or inter-organizational R&D cooperation seems to be fruitful. Even if it is considered that there are regions to be found where the actors of the regional innovation system are well linked (Günther, 2004), there still may be an argument for such a policy. Based on the evolutionary approaches and resource-based views of the growth of the firm, it was found that the firms in an advanced stage of development tend to enter alliances and innovation networks of higher functionality (Ahuja 2000; Overby, 2005; and Lewrick *et al.*, 2007) because stronger resources search and attract stronger resources. Thus, reasoning that 'network policy' would be less effective up to a certain degree of inter-organizational linkages (Günther, 2005) may be misleading if the quality of the links

is ignored. In this study, for example, 60% of the organizations exhibited R&D cooperation experience, only 30% of the organizations were formerly involved in regional networks of innovation.

With respect to innovation policy, the findings relating to the universities' knowledge transfer and their 'antenna function' support the general approach that stimulates the integration of public research organizations into regional innovation networks. But as the results indicate the frequency of inter-regional R&D cooperation activities has to be taken into account as well. The more frequent the universities conduct interregional R&D cooperation (in product innovation as well as process innovations), the more knowledge they transfer to network partners. From these findings, there results three implications:

1. Public research organizations that exhibit rich inter-regional R&D cooperation should be integrated into policy programs;
2. First of all, this should be to those public research organizations that conduct inter-regional R&D cooperation in product as well as process innovations; and
3. The policy should be addressed to the initial stages of development of inter-regional R&D cooperation too.

If a distinction has been made between certain groups following the regional system of innovation approach, it is above all universities that transfer knowledge to their network partners. However, the service firms have also shown themselves to be the partners with whom collaboration involves an extent of knowledge exchange above average. These results correspond to the findings that knowledge intensive service firms can be seen as important medium for knowledge spillovers (Leydesdorff and Fritsch, 2005). As the results further indicate, a great extent of the knowledge transferred by the universities or by the other groups of actors arrives at the firms. With respect to the manufacturing firms of the sample that predominantly attained 'net-benefits' in the course of knowledge exchange, the results show that regional innovation networks, in principle, operate according to ideal and typical conceptions. However, there are results that seem somewhat surprising. The non-university public research organizations exhibit a relatively low extent of knowledge transfer. As well, they do not involve an 'antenna function'. One reason for this may be the following: While the universities tend to locate at central places, some of the non-university research organizations are located at more peripheral regions that probably lack firms with adequate absorptive capacity. By all means, the results indicate that the firms' absorptive capacity matters with regard to the extent of knowledge they were able to acquire.

In principle, the findings can be regarded as being transferable from regional innovation networks to regional innovation systems. This applies in particular to the behavior-conditioned results. With regard to the effectiveness of the 'antenna function' of public research, however, it may rather be that the findings are transferable to those regions with similar socioeconomic characteristics, the reason being that this antenna function

conceivably becomes effective where firms engage in less frequent inter-regional R&D cooperation. ■

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Appendix

List of Variables		
Variables	Description	Measurement
KNOW	Knowledge acquisition/absorption Queried as: "Technical support received from network partners".	1 = not at all 2 = very few 3 = moderate 4 = much 5 = very much
NW-EXP	Regional innovation network experience.	1 = yes; 2 = no
CO_EXPBR	Cooperation experience in basic research.	1 = not at all
CO_EXPPDI _{Inno}	Cooperation experience in product innovation.	2 = rarely
CO_EXPPCI _{Inno}	Cooperation experience in process innovation.	3 = from time-to-time 4 = often 5 = very often
CON_INT	Degree of the partners' intensity of contact.	1 = not at all
TRUST	Degree of fairness and trust between the partners.	3 = partly
COM_NORMS	Degree of common norms and values.	5 = completely
IDENT	Degree of identification with the network.	
FIRM_SIZE	Size of the firms.	1 = up to 10 employees 2 = 11 up to 50 3 = 51 up to 100 4 = 101 up to 250 5 = 251 and above
Acquired information/ knowledge	Queried as: "Information/technical support received from network partners."	1 = not at all 2 = very few 3 = moderate
Transferred information/ knowledge	Queried as: "I was helping partners by information/technical support".	4 = much 5 = very much
Inter-regional cooperation frequency	Queried as: "How often do you cooperate on basic research/product innovation/process innovation?"	1 = not at all 2 = rarely 3 = from time to time 4 = often 5 = very often
Degree of specialization	Degree of specialization that results from the cooperation activity within the network.	1 = not at all 2 = low 3 = moderate 4 = high 5 = very high

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